

The Soul of Wine

Data Appendix

Source Data, Tables, and Reference Materials

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Companion documents: Summary · Technical Appendix · Methods Primer · Data Appendix · Region Reference (Identity, Terroir)

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D.1 Region Roster

The 59 regions analysed in this study, grouped by country of primary regulatory affiliation. Old World total: 39 regions across 10 countries. New World total: 20 regions across 6 countries.

Country	Regions
<i>Old World</i>	
Austria	Kamptal, Steiermark, Wachau, Wagram
Croatia	Dalmatian Coast
France	Alsace, Beaujolais, Bordeaux, Burgundy, Champagne, Châteauneuf-du-Pape, Jura, Loire, Northern Rhône, Provence
Germany	Baden, Mosel, Nahe, Pfalz, Rheingau
Greece	Macedonia, Santorini
Hungary	Tokaj
Italy	Alto Adige, Campania, Etna, Friuli-Venezia Giulia, Liguria, Piedmont, Sardinia, Sicily, Tuscany, Veneto
Portugal	Douro
Slovenia	Goriška Brda
Spain	Catalonia, Galicia, Ribera del Duero, Rioja
<i>New World</i>	
Argentina	Mendoza, Patagonia
Australia	Barossa Valley, Hunter Valley, Margaret River
Chile	Maipo Valley
New Zealand	Central Otago, Hawke's Bay, Marlborough
S. Africa	Stellenbosch, Swartland
USA	Columbia Valley, Finger Lakes, Napa Valley, Paso Robles, Santa Barbara, Santa Cruz Mountains, Sonoma, Walla Walla, Willamette Valley

D.2 D-Score Matrix and Identity Cluster Membership

All 59 regions with the six identity D-scores (D_1 – D_6 on -2 to $+2$ scale), the assigned identity cluster, and the canonical soul metaphor. Sorted by identity cluster, then by region number within each cluster. Cluster definitions: D_1 Interiority $\leftrightarrow$ Exteriority; D_2 Struggle $\leftrightarrow$ Ease; D_3 Tradition $\leftrightarrow$ Reinvention; D_4 Individual $\leftrightarrow$ Collective; D_5 Urgency $\leftrightarrow$ Timelessness; D_6 Earthly $\leftrightarrow$ Transcendent.

#	Region	Country	Metaphor	D_1	D_2	D_3	D_4	D_5	D_6
Old World Interior (10 regions)									
9	Burgundy	France	Devotion	+2	+1	+2	+1	−1	−1
25	Campania	Italy	Memory	+1	0	+1	0	−1	−1
26	Etna	Italy	Awakening	+1	+2	0	0	+1	−1
37	Galicia	Spain	Longing (Morriña)	+1	+1	+1	+1	0	−1
17	Mosel	Germany	Poetry	+2	+2	+2	+1	−2	−2
14	Northern Rhône	France	Solitude	+2	+2	+2	+2	−2	−1
29	Piedmont	Italy	Philosophy	+2	+1	+2	+1	−2	−1
20	Rheingau	Germany	Nobility	+1	0	+2	+1	−2	0
56	Santa Cruz Mtns	USA	Obsession	+2	+1	0	+2	0	0
23	Tokaj	Hungary	Melancholy	+2	+1	+2	+1	−2	−1
Outward Ease (5 regions)									
7	Beaujolais	France	Joy	−1	−1	−1	−1	+1	+1
48	Marlborough	NZ	Assertion	−2	0	−1	−1	+2	+1
15	Provence	France	Pleasure	−2	−2	0	−1	+1	+1
33	Veneto	Italy	Commerce	−2	0	−1	−1	+1	+1
58	Walla Walla	USA	Community	+1	−1	−1	−2	+1	+1
New World Reinvention (12 regions)									
36	Catalonia	Spain	Identity	−1	+1	−2	+1	+2	+1
46	Central Otago	NZ	Adventure	−1	+1	−1	+1	+1	0
51	Columbia Valley	USA	Determination	−1	+1	−1	+1	+1	+2
52	Finger Lakes	USA	Conviction	−1	+1	−1	+1	+1	0
40	Mendoza	Argentina	Reinvention	−1	0	−2	0	+1	+1
53	Napa Valley	USA	Ambition	−2	0	−1	−1	+2	+1
54	Paso Robles	USA	Independence	−1	0	−2	+1	+1	+1
41	Patagonia	Argentina	Extremity	0	+2	−2	+1	+2	0

#	Region	Country	Metaphor	D_1	D_2	D_3	D_4	D_5	D_6
31	Sicily	Italy	Resurrection	-1	+1	-2	+1	+1	+1
49	Stellenbosch	S. Africa	Aspiration	-1	0	-1	+1	+1	+1
50	Swartland	S. Africa	Rebellion	-1	+1	-2	+1	+2	+1
57	Sonoma	USA	Authenticity	-1	0	-1	+1	0	+1
Old World Exterior (7 regions)									
16	Baden	Germany	Warmth	-1	-1	+1	-1	-1	+1
8	Bordeaux	France	Business	-2	-1	+2	-2	0	+1
10	Champagne	France	Society	-2	-1	+2	-2	0	-1
11	Châteauneuf-du-Pape	France	Family	-1	0	+2	-2	-1	+1
5	Dalmatian Coast	Croatia	Tranquility	+2	-2	+1	-1	-2	+2
19	Pfalz	Germany	Generosity	-1	-2	+1	-1	-1	+1
39	Rioja	Spain	Patience	-1	0	+2	-1	-1	+1
Against the Odds (10 regions)									
42	Barossa Valley	Australia	Fortitude	+1	+2	+1	0	0	+1
34	Douro	Portugal	Endurance	+1	+2	+2	+1	-1	+1
43	Hunter Valley	Australia	Defiance	+1	+2	+1	+1	-1	+1
12	Jura	France	Eccentricity	+2	+1	+2	+1	0	+1
28	Liguria	Italy	Intimacy	+2	+1	+1	+1	-1	+1
21	Macedonia	Greece	Austerity	+2	+2	+2	+1	0	+1
38	Ribera del Duero	Spain	Severity	+1	+2	+1	+1	0	+2
22	Santorini	Greece	Survival	+2	+2	+1	+1	+1	+1
30	Sardinia	Italy	Stubbornness	+2	+1	+2	-1	-1	+2
4	Wagram	Austria	Earth	+2	0	+2	0	0	+1
The Moderates (15 regions)									
6	Alsace	France	Duality	0	0	+1	0	0	0
24	Alto Adige	Italy	Precision	0	0	+1	0	0	+1
27	Friuli-VG	Italy	Dialogue	-1	0	0	0	0	0
35	Goriška Brda	Slovenia	Fortune	-1	0	0	0	0	+1
47	Hawke's Bay	NZ	Confidence	0	0	0	0	-1	+1
1	Kamptal	Austria	Discipline	+1	0	+1	+1	0	0
13	Loire	France	Sentimentality	-1	0	0	0	0	0
45	Maipo Valley	Chile	Pride	-1	0	+1	0	-1	+1
44	Margaret River	Australia	Composure	0	0	0	0	-1	0

#	Region	Country	Metaphor	D_1	D_2	D_3	D_4	D_5	D_6
18	Nahe	Germany	Subtlety	+1	0	+1	+1	-1	0
55	Santa Barbara	USA	Serendipity	-1	0	0	0	0	+1
2	Steiermark	Austria	Clarity	+1	0	0	+1	+1	+1
32	Tuscany	Italy	Art	-1	-1	0	+1	0	0
3	Wachau	Austria	Monumentality	0	0	+1	+1	0	+1
59	Willamette Valley	USA	Idealism	-1	+1	-1	+1	0	-1

D.3 Terroir Cluster Membership

The terroir partition is fully model-derived: Layer 2 TF-IDF $\rightarrow$ PCA (10 components) $\rightarrow$ K-means ($k = 7$). Cluster count, membership, and naming are determined by the algorithm with no manual override. Top TF-IDF terms indicate the vocabulary features that most distinguish each cluster.

Iberian Continental (6). Top terms: *spain, rioja, do.*

Douro, Catalonia, Galicia, Ribera del Duero, Rioja, Mendoza.

Southern Hemisphere & International (12). Top terms: *zealand, new zealand, australia.*

Alto Adige, Patagonia, Barossa Valley, Hunter Valley, Margaret River, Maipo Valley, Central Otago, Hawke’s Bay, Marlborough, Stellenbosch, Swartland, Napa Valley.

American West Coast (7). Top terms: *ava, wineries, california.*

Columbia Valley, Paso Robles, Santa Barbara, Santa Cruz Mountains, Sonoma, Walla Walla, Willamette Valley.

French Viticultural (10). Top terms: *france, côte, de.*

Alsace, Beaujolais, Bordeaux, Burgundy, Champagne, Châteauneuf-du-Pape, Jura, Loire, Northern Rhône, Provence.

Germanic Rhine (5). Top terms: *germany, vdp, mosel.*

Baden, Mosel, Nahe, Pfalz, Rheingau.

Austrian Danube (3). Top terms: *dac, veltliner, austria.*

Kamptal, Wachau, Wagram.

Mediterranean & Volcanic (16). Top terms: *italy, di, indigenous.*

Steiermark, Dalmatian Coast, Macedonia, Santorini, Tokaj, Campania, Etna, Friuli-Venezia Giulia, Liguria, Piedmont, Sardinia, Sicily, Tuscany, Veneto, Goriška Brda, Finger Lakes.

D.4 Identity $\times$ Terroir Contingency Table

The full 6×7 contingency table $\{n_{ij}\}$ counting regions in each (identity, terroir) cell. Row totals a_i are the identity-cluster sizes; column totals b_j are the terroir-cluster sizes; the grand total is $n = 59$.

	Iber.	S.Hem.	Am.W.	Fre.	Ger.	Aus.	Med.	a_i
Old World Interior	1	0	1	1	1	0	6	10
Outward Ease	0	1	1	2	0	0	1	5
New World Reinvention	2	5	1	0	0	0	4	12
Old World Exterior	1	0	0	3	2	0	1	7
Against the Odds	2	2	0	1	0	1	4	10
The Moderates	0	4	4	3	2	2	0	15
b_j	6	12	7	10	5	3	16	59

Test statistics on this table: $\chi^2 = 38.776$, $df = 30$, $p = 0.131$. ARI = 0.023.

D.5 Per-Cluster Silhouette Breakdown

Mean silhouette score per cluster, with cluster size in parentheses. The overall identity silhouette is $S_I = 0.3029$; the overall terroir silhouette is $S_T = 0.2457$.

<i>Identity</i>		<i>Terroir</i>	
Cluster (n)	Mean s	Cluster (n)	Mean s
Old World Interior (10)	0.34	Iberian Continental (6)	0.27
Outward Ease (5)	0.31	Southern Hem. & International (12)	0.21
New World Reinvention (12)	0.32	American West Coast (7)	0.26
Old World Exterior (7)	0.27	French Viticultural (10)	0.28
Against the Odds (10)	0.33	Germanic Rhine (5)	0.31
The Moderates (15)	0.25	Austrian Danube (3)	0.29
		Mediterranean & Volcanic (16)	0.20

The Moderates is the weakest identity cluster ($s = 0.25$), reflecting its definitional character as the cluster of pluralistic regions where no single identity force dominates. Mediterranean & Volcanic is the weakest terroir cluster ($s = 0.20$), reflecting its size (16 regions) and substantive heterogeneity (Etna and Santorini volcanic; Tuscany and Piedmont continental Mediterranean; Finger Lakes a New World outlier sharing terroir vocabulary with the Mediterranean group).

D.6 SME Change Log

Sixteen score and metaphor changes across thirteen regions, applied after subject-matter expert review of the Pass 5 D-score matrix. All changes are final and incorporated into the canonical documents. Three intentional cluster shifts resulted: Etna and Santa Cruz Mountains → Old World Interior; Nahe and Wachau → The Moderates; Sonoma → New World Reinvention.

Region	Metaphor	Change	Rationale
Bordeaux	Business	$D_3: +1 \rightarrow +2$	Deeper institution separates from the region.
Champagne	Society	$D_3: +1 \rightarrow +2$	Matches Bordeaux as dominant institution.
Etna	Awakening	$D_1: -1 \rightarrow +1, D_2: +1 \rightarrow +2, D_3: -1 \rightarrow 0$	Volcanic introduction of extreme conditions. Cluster shift: $-1 \rightarrow 0$.
Goriška Brda	Fortune	Metaphor: Serendipity $\rightarrow$ Fortune; $D_3: -1 \rightarrow 0$	Resolves duplication of Barbara; accidental intervention.
Rioja	Patience	Metaphor: Memory $\rightarrow$ Patience	Resolves duplication of patience captures opportunity.
Santa Barbara	Serendipity	$D_3: -1 \rightarrow 0$	Serendipity is doubly balanced on trajectory.
Santa Cruz Mtns	Obsession	$D_2: +2 \rightarrow +1$	Obsessive private endurance. Cluster shift: Interior.
Sonoma	Authenticity	$D_1: 0 \rightarrow -1, D_3: 0 \rightarrow -1$	Deliberate introduction of intent; modification. Cluster shift: $-1 \rightarrow 0$.
Stellenbosch	Aspiration	$D_4: 0 \rightarrow +1$	Individual professional anchors in New World.
Tuscany	Art	Metaphor: Heritage $\rightarrow$ Art; $D_2: 0 \rightarrow -1, D_3: +1 \rightarrow 0$	Art better captures cultural frame; modernized tradition.
Wachau	Monumentality	$D_1: +1 \rightarrow 0, D_2: +1 \rightarrow 0$	Balanced, complex extreme struggle. Moderates.
Wagram	Earth	$D_1: +1 \rightarrow +2, D_3: +1 \rightarrow +2, D_6: +2 \rightarrow +1$	Deeply earthly custodial tradition. scape.
Walla Walla	Community	$D_1: 0 \rightarrow +1$	Place-anchored inward-facing development reach.

References

- Hubert, L. & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1), 193–218.
- Pearson, K. (1900). On the criterion that a given system of deviations from the probable. *Philosophical Magazine*, 50(302), 157–175.
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.